

Experiment 2

SQL Queries, Sub-queries and Aggregate Functions
Sailors Database | Oracle Database 21c XE

1. Source Code

```
-- -----
-- Database creation (DDL and DML)
-- -----

-- 2-1. Create the tables of the Sailors database
CREATE TABLE sailors (
  sid      NUMBER(3) PRIMARY KEY,
  sname    VARCHAR2(20) NOT NULL,
  rating   NUMBER(2),
  age      NUMBER(4,1)
);
CREATE TABLE boats (
  bid      NUMBER(3) PRIMARY KEY,
  bname    VARCHAR2(20),
  color    VARCHAR2(10)
);
CREATE TABLE reserves (
  sid      NUMBER(3) REFERENCES sailors(sid),
  bid      NUMBER(3) REFERENCES boats(bid),
  day      DATE,
  PRIMARY KEY (sid, bid, day)
);

-- 2-2. Insert the sample data
INSERT INTO sailors VALUES (22, 'Dustin', 7, 45.0);
INSERT INTO sailors VALUES (29, 'Brutus', 1, 33.0);
INSERT INTO sailors VALUES (31, 'Lubber', 8, 55.5);
INSERT INTO sailors VALUES (32, 'Andy', 8, 25.5);
INSERT INTO sailors VALUES (58, 'Rusty', 10, 35.0);
INSERT INTO sailors VALUES (64, 'Horatio', 7, 35.0);
INSERT INTO sailors VALUES (71, 'Zorba', 10, 16.0);
INSERT INTO sailors VALUES (74, 'Horatio', 9, 35.0);
INSERT INTO sailors VALUES (85, 'Art', 3, 25.5);
INSERT INTO sailors VALUES (95, 'Bob', 3, 63.5);
INSERT INTO boats VALUES (101, 'Interlake', 'blue');
INSERT INTO boats VALUES (102, 'Interlake', 'red');
INSERT INTO boats VALUES (103, 'Clipper', 'green');
INSERT INTO boats VALUES (104, 'Marine', 'red');
INSERT INTO reserves VALUES (22, 101, TO_DATE('10-10-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (22, 102, TO_DATE('10-10-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (22, 103, TO_DATE('08-10-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (22, 104, TO_DATE('07-10-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (31, 102, TO_DATE('10-11-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (31, 103, TO_DATE('06-11-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (31, 104, TO_DATE('12-11-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (64, 101, TO_DATE('05-09-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (64, 102, TO_DATE('08-09-1998', 'DD-MM-YYYY'));
INSERT INTO reserves VALUES (74, 103, TO_DATE('08-09-1998', 'DD-MM-YYYY'));
COMMIT;

-- 2-3. Display the tables
SELECT * FROM sailors;
SELECT * FROM boats;
SELECT * FROM reserves;

-- -----
-- 2a. Queries and sub-queries (ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, MINUS)
-- -----

-- Q1. Names and ages of all sailors
SELECT sname, age FROM sailors;

-- Q2. Sailors with a rating above 7
SELECT * FROM sailors WHERE rating > 7;

-- Q3. Names of sailors who have reserved boat 103
SELECT s.sname
FROM sailors s, reserves r
```

```

WHERE s.sid = r.sid AND r.bid = 103;

-- Q4. Sids of sailors who have reserved a red boat
SELECT DISTINCT r.sid
FROM reserves r, boats b
WHERE r.bid = b.bid AND b.color = 'red';

-- Q5. Names of sailors who have reserved a red boat
SELECT DISTINCT s.sname
FROM sailors s, reserves r, boats b
WHERE s.sid = r.sid AND r.bid = b.bid AND b.color = 'red';

-- Q6. Colors of boats reserved by Lubber
SELECT DISTINCT b.color
FROM sailors s, reserves r, boats b
WHERE s.sid = r.sid AND r.bid = b.bid AND s.sname = 'Lubber';

-- Q7. Names of sailors who have reserved at least one boat
SELECT DISTINCT s.sname
FROM sailors s, reserves r
WHERE s.sid = r.sid;

-- Q8. Rating increments of sailors who sailed two different boats on the same day
SELECT DISTINCT s.sname, s.rating + 1 AS rating
FROM sailors s, reserves r1, reserves r2
WHERE s.sid = r1.sid AND s.sid = r2.sid
    AND r1.day = r2.day AND r1.bid <> r2.bid;

-- Q9. Ages of sailors whose name begins and ends with B and has at least three characters
SELECT age
FROM sailors
WHERE UPPER(sname) LIKE 'B_%B';

-- Q10. Names of sailors who reserved a red boat or a green boat (UNION)
SELECT s.sname FROM sailors s, reserves r, boats b
WHERE s.sid = r.sid AND r.bid = b.bid AND b.color = 'red'
UNION
SELECT s.sname FROM sailors s, reserves r, boats b
WHERE s.sid = r.sid AND r.bid = b.bid AND b.color = 'green';

-- Q11. Names of sailors who have reserved both a red and a green boat (INTERSECT)
SELECT sname FROM sailors
WHERE sid IN (SELECT r.sid FROM reserves r, boats b WHERE r.bid = b.bid AND b.color = 'red'
    INTERSECT
    SELECT r.sid FROM reserves r, boats b WHERE r.bid = b.bid AND b.color = 'green');

-- Q12. Sids of sailors who reserved red boats but not green boats (MINUS)
SELECT r.sid FROM reserves r, boats b WHERE r.bid = b.bid AND b.color = 'red'
MINUS
SELECT r.sid FROM reserves r, boats b WHERE r.bid = b.bid AND b.color = 'green';

-- Q13. Sids of sailors who have a rating of 10 or have reserved boat 104
SELECT sid FROM sailors WHERE rating = 10
UNION
SELECT sid FROM reserves WHERE bid = 104;

-- Q14. Names of sailors who have reserved boat 103 (IN)
SELECT sname FROM sailors
WHERE sid IN (SELECT sid FROM reserves WHERE bid = 103);

-- Q15. Names of sailors who have reserved a red boat (nested IN)
SELECT sname FROM sailors
WHERE sid IN (SELECT sid FROM reserves
    WHERE bid IN (SELECT bid FROM boats WHERE color = 'red'));

-- Q16. Names of sailors who have reserved boat number 103 (EXISTS)
SELECT s.sname FROM sailors s
WHERE EXISTS (SELECT * FROM reserves r
    WHERE r.sid = s.sid AND r.bid = 103);

-- Q17. Sailors whose rating is better than some sailor called Horatio (ANY)
SELECT * FROM sailors
WHERE rating > ANY (SELECT rating FROM sailors WHERE sname = 'Horatio');

-- Q18. Sailors whose rating is better than every sailor called Horatio (ALL)
SELECT * FROM sailors
WHERE rating > ALL (SELECT rating FROM sailors WHERE sname = 'Horatio');
-- Q19. Sailors with the highest rating (ALL)

```

```

SELECT * FROM sailors
WHERE rating >= ALL (SELECT rating FROM sailors);

-- Q20. Names of sailors who have reserved both a red and a green boat (IN ... AND IN)
SELECT sname FROM sailors
WHERE sid IN (SELECT r.sid FROM reserves r, boats b WHERE r.bid = b.bid AND b.color = 'red')
  AND sid IN (SELECT r.sid FROM reserves r, boats b WHERE r.bid = b.bid AND b.color = 'green');

-- Q21. Names of sailors who have reserved all boats (NOT EXISTS)
SELECT s.sname FROM sailors s
WHERE NOT EXISTS (SELECT b.bid FROM boats b
  WHERE NOT EXISTS (SELECT * FROM reserves r
    WHERE r.bid = b.bid AND r.sid = s.sid));

-----
-- 2b. Aggregate functions, GROUP BY and HAVING
-----

-- Q22. Average age of all sailors
SELECT AVG(age) FROM sailors;

-- Q23. Average age of sailors with a rating of 10
SELECT AVG(age) FROM sailors WHERE rating = 10;

-- Q24. Name and age of the oldest sailor
SELECT sname, age FROM sailors WHERE age = (SELECT MAX(age) FROM sailors);

-- Q25. Count the number of sailors
SELECT COUNT(*) FROM sailors;

-- Q26. Count the number of different sailor names
SELECT COUNT(DISTINCT sname) FROM sailors;

-- Q27. Names of sailors older than the oldest sailor with a rating of 10
SELECT sname FROM sailors
WHERE age > (SELECT MAX(age) FROM sailors WHERE rating = 10);

-- Q28. Age of the youngest sailor for each rating level
SELECT rating, MIN(age) FROM sailors
GROUP BY rating ORDER BY rating;

-- Q29. Youngest sailor of voting age for each rating level with at least two such sailors
SELECT rating, MIN(age) AS min_age
FROM sailors
WHERE age >= 18
GROUP BY rating
HAVING COUNT(*) >= 2;

-- Q30. For each red boat, the number of reservations
SELECT b.bid, COUNT(*) AS reservation_count
FROM boats b, reserves r
WHERE r.bid = b.bid AND b.color = 'red'
GROUP BY b.bid;

-- Q31. Average age of sailors for each rating level that has at least two sailors
SELECT rating, AVG(age)
FROM sailors
GROUP BY rating
HAVING COUNT(*) >= 2;

-- Q32. Average age of voting-age sailors for each rating level that has at least two sailors
SELECT s.rating, AVG(s.age)
FROM sailors s
WHERE s.age >= 18
GROUP BY s.rating
HAVING 1 < (SELECT COUNT(*) FROM sailors s2
  WHERE s2.rating = s.rating);

-- Q33. Average age of voting-age sailors for each rating level that has at least two such sailors
SELECT rating, AVG(age)
FROM sailors
WHERE age >= 18
GROUP BY rating
HAVING COUNT(*) >= 2;

-- Q34. Ratings for which the average age is the minimum over all ratings
SELECT rating, AVG(age)
FROM sailors

```

```
GROUP BY rating
HAVING AVG(age) = (SELECT MIN(AVG(age)) FROM sailors GROUP BY rating);
```

```
-- -----
-- MNC interview questions
-- -----
```

```
-- MNC-1. Second highest rating
SELECT MAX(rating) FROM sailors
WHERE rating < (SELECT MAX(rating) FROM sailors);
```

```
-- MNC-2. Names of sailors who have never reserved a boat
SELECT sname FROM sailors
WHERE sid NOT IN (SELECT sid FROM reserves);
```

2. Execution and Output

Output shown is sample output of the programs above (Oracle Database 21c XE, SQL*Plus).

Database creation (DDL and DML)

```
[2-1]
Table created.
Table created.
Table created.
```

[illegible]

[2-3]

SID	SNAME	RATING	AGE
22	Dustin	7	45
29	Brutus	1	33
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35
64	Horatio	7	35
71	Zorba	10	16
74	Horatio	9	35
85	Art	3	25.5
95	Bob	3	63.5

```
10 rows selected.
```

BID	BNAME	COLOR
101	Interlake	blue
102	Interlake	red
103	Clipper	green
104	Marine	red

```
4 rows selected.
```

SID	BID	DAY
22	101	10-OCT-98
22	102	10-OCT-98
22	103	08-OCT-98
22	104	07-OCT-98
31	102	10-NOV-98
31	103	06-NOV-98
31	104	12-NOV-98
64	101	05-SEP-98
64	102	08-SEP-98
74	103	08-SEP-98

```
10 rows selected.
```

2a. Queries and sub-queries (ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, MINUS)

[Q1]

SNAME	AGE
Dustin	45
Brutus	33
Lubber	55.5
Andy	25.5
Rusty	35
Horatio	35
Zorba	16
Horatio	35
Art	25.5
Bob	63.5

10 rows selected.

[Q2]

SID	SNAME	RATING	AGE
31	Lubber	8	55.5
32	Andy	8	25.5
58	Rusty	10	35
71	Zorba	10	16
74	Horatio	9	35

5 rows selected.

[Q3]

SNAME
Dustin
Lubber
Horatio

3 rows selected.

[Q4]

SID
22
31
64

3 rows selected.

[Q5]

SNAME
Dustin
Lubber
Horatio

3 rows selected.

[Q6]

COLOR
red
green

2 rows selected.

[Q7]

SNAME
Dustin
Lubber
Horatio

3 rows selected.

[Q8]

SNAME	RATING
-------	--------

```
-----
Dustin      8
```

1 row selected.

[Q9]

```
AGE
----
63.5
```

1 row selected.

[Q10]

```
SNAME
-----
Dustin
Horatio
Lubber
```

3 rows selected.

[Q11]

```
SNAME
-----
Dustin
Lubber
```

2 rows selected.

[Q12]

```
SID
---
64
```

1 row selected.

[Q13]

```
SID
---
22
31
58
71
```

4 rows selected.

[Q14]

```
SNAME
-----
Dustin
Lubber
Horatio
```

3 rows selected.

[Q15]

```
SNAME
-----
Dustin
Lubber
Horatio
```

3 rows selected.

[Q16]

```
SNAME
-----
Dustin
Lubber
Horatio
```

3 rows selected.

[Q17]

```
SID SNAME  RATING  AGE
---
-----
```

```

31 Lubber      8 55.5
32 Andy        8 25.5
58 Rusty      10 35
71 Zorba      10 16
74 Horatio    9 35

```

5 rows selected.

[Q18]

```

SID SNAME RATING AGE
---
58 Rusty    10 35
71 Zorba    10 16

```

2 rows selected.

[Q19]

```

SID SNAME RATING AGE
---
58 Rusty    10 35
71 Zorba    10 16

```

2 rows selected.

[Q20]

```

SNAME
-----
Dustin
Lubber

```

2 rows selected.

[Q21]

```

SNAME
-----
Dustin

```

1 row selected.

2b. Aggregate functions, GROUP BY and HAVING

[Q22]

```

AVG(AGE)
-----
      36.9

```

1 row selected.

[Q23]

```

AVG(AGE)
-----
      25.5

```

1 row selected.

[Q24]

```

SNAME  AGE
-----
Bob    63.5

```

1 row selected.

[Q25]

```

COUNT(*)
-----
      10

```

1 row selected.

[Q26]

```

COUNT(DISTINCT SNAME)
-----
                        9

```

1 row selected.

[Q27]

```
SNAME
-----
Dustin
Lubber
Bob
```

3 rows selected.

[Q28]

```
RATING MIN(AGE)
-----
      1      33
      3     25.5
      7      35
      8     25.5
      9      35
     10      16
```

6 rows selected.

[Q29]

```
RATING MIN_AGE
-----
      3     25.5
      7      35
      8     25.5
```

3 rows selected.

[Q30]

```
BID RESERVATION_COUNT
---
102                3
104                2
```

2 rows selected.

[Q31]

```
RATING AVG(AGE)
-----
      3     44.5
      7      40
      8     40.5
     10     25.5
```

4 rows selected.

[Q32]

```
RATING AVG(S.AGE)
-----
      3     44.5
      7      40
      8     40.5
     10      35
```

4 rows selected.

[Q33]

```
RATING AVG(AGE)
-----
      3     44.5
      7      40
      8     40.5
```

3 rows selected.

[Q34]

```
RATING AVG(AGE)
-----
     10     25.5
```

1 row selected.

MNC interview questions

[MNC-1]

MAX(RATING)

9

1 row selected.

[MNC-2]

SNAME

Brutus

Andy

Rusty

Zorba

Art

Bob

6 rows selected.